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Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

AUG 22 2019

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388/2019-002-00
UNIT 2 LICENSE NO. NPF-22
PLA-7807

Docket No. 50-388

Attached is Licensee Event Report (LER) 50-388/2019-002-00. This LER is reporting a condition in which there was a loss of the Unit 2 B Reactor Protection System Bus, which resulted in a valid actuation of the Primary Containment Isolation System. This condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as a valid actuation of a system identified in 10 CFR 50.73(a)(2)(iv)(B). There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to be "K. Cimorelli", written over a horizontal line.

K. Cimorelli

Attachment: LER 50-388/2019-002-00

Copy: NRC Region I
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Ms. T. E. Hood, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

NRC FORM 366 (04-2018)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020								
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)											
1. Facility Name Susquehanna Steam Electric Station Unit 2					2. Docket Number 05000388		3. Page 1 OF 3						
4. Title Valid Actuation of the Primary Containment Isolation System due to a Trip of Reactor Protection System Bus 2B													
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved				
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number			
06	27	2019	2019	002	00	08	22	2019		05000			
									Facility Name	Docket Number			
										05000			
9. Operating Mode 1		11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
		☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)			☐ 50.73(a)(2)(viii)(A)		
		☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)			☐ 50.73(a)(2)(viii)(B)		
		☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)			☐ 50.73(a)(2)(ix)(A)		
		☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☒ 50.73(a)(2)(iv)(A)			☐ 50.73(a)(2)(x)		
10. Power Level 100		☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(v)(A)			☐ 73.71(a)(4)		
		☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☐ 50.73(a)(2)(v)(B)			☐ 73.71(a)(5)		
		☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☐ 50.73(a)(2)(v)(C)			☐ 73.77(a)(1)		
		☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☐ 50.73(a)(2)(v)(D)			☐ 73.77(a)(2)(ii)		
		☐ 20.2203(a)(2)(vi)			☐ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)			☐ 73.77(a)(2)(iii)		
					☐ 50.73(a)(2)(i)(C)			☐ Other (Specify in Abstract below or in NRC Form 366A					
12. Licensee Contact for this LER													
Licensee Contact Shane Jurek, Senior Engineer – Nuclear Regulatory Affairs								Telephone Number (Include Area Code) (570) 542-1695					
13. Complete One Line for each Component Failure Described in this Report													
Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES				
X	JC	RLY	G080	Y									
14. Supplemental Report Expected ☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No					15. Expected Submission Date			Month	Day	Year			
Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines) On June 27, 2019, at 0937 a half scram signal was received on Unit 2, Division 2 of the Reactor Protection System (RPS). The half scram signal resulted in a trip of the 2B RPS instrument bus, a trip of the 2A Residual Heat Removal Service Water (RHRSW) Pump, a loss of the 2A and 2B Containment Radiation Monitors (CRMs), and Primary Containment Isolation System (PCIS) actuations. The loss of both CRMs resulted in entering Condition B of Technical Specifications (TS) Limiting Condition for Operation (LCO) 3.4.6, "RCS Leakage Detection Instrumentation." The PCIS actuation resulted in isolation of containment isolation valves in multiple systems and was determined to be reportable in accordance with 10 CFR 50.72(b)(3)(iv)(A) for the valid isolation signal. The 2B RPS instrument bus was transferred to its alternate power supply and the 2A and 2B CRMs were restored. The 2B RPS instrument bus was realigned to its normal source on July 2, 2019. The condition was also determined to be reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) for the valid PCIS signal. The event was determined to be caused by an age-related failure of a control relay on the normal power supply to the 2B RPS instrument bus. Corrective actions included replacing the failed relay. There were no actual safety consequences associated with the condition.													

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station Unit 2	05000-388	2019	002	00

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On June 27, 2019, at approximately 0937, a half scram signal was received on Susquehanna Steam Electric Station (SSES) Unit 2, Division 2 of the RPS [EIS Code: JC]. Concurrently, operators were performing SO-216-A03, "Quarterly Residual Heat Removal Service Water (RHRSW) System Flow Verification Division I," The half scram signal resulted in a trip of the 2B RPS instrument bus, a trip of the 2A RHRSW Pump [EIS System / Component Code: BO/P], a loss of the 2A and 2B CRMs [EIS System / Component Code: IL / MON], and PCIS [EIS Code: JM] actuations. The loss of both CRMs resulted in entering Condition B of SSES TS LCO 3.4.6, "RCS Leakage Detection Instrumentation." The 2B RPS instrument bus was transferred to its alternate power supply at 0941 and the 2A and 2B CRMs were restored at 0956 and 0959, respectively; Condition B of LCO 3.4.6 was exited at 1000; all TS Completion Times were met.

Upon tripping the 2B RPS instrument bus, SSES personnel performed troubleshooting on the normal power supply to the bus. It was determined that Control Relay 2Y201BC72BK1B [Component Code: RLY] (hereafter referred to as the K1B relay) failed. The K1B relay was replaced and SSES personnel restored the 2B RPS instrument bus to its normal source on July 2, 2019.

The PCIS actuations resulted in isolations of multiple systems as designed. As a result, this condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in automatic actuation of a system as identified in 10 CFR 50.73(a)(2)(iv)(B)(2).

CAUSE OF EVENT

The direct cause of the event was determined to be an age-related failure of the K1B relay.

ANALYSIS/SAFETY SIGNIFICANCE

Failure of the K1B relay caused a loss of power to one of two power distribution systems of the 'B' RPS. The RPS is "one out of two taken twice" logic. The loss of power resulted in a valid isolation signal on the 'B' logic division. This response is consistent with the safety analysis. Since the plant response was as designed, all accident functions would have been met. Therefore, there was no loss of safety function. Accordingly, this event will not be counted as a safety system functional failure in the Reactor Oversight Process Performance Indicators.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 2		05000-388	YEAR	SEQUENTIAL NUMBER	REV NO.
			2019	002	00

There were no actual consequences to the health and safety of the public as a result of this event.

CORRECTIVE ACTIONS

Corrective actions included replacement of the K1B relay.

COMPONENT FAILURE INFORMATION

Component Identification – 2Y201BC72BK1B

Component Name – B RPS Normal Power Contactor

Component Manufacturer – GE

Relay Type – GE Type CR305 Relay

Manufacturer Part Number – 117C2491P002

PREVIOUS OCCURRENCES

LER 50-388(387)/2000-005-01, “ESF Actuations Due to RPS EPA Breaker Trip,” dated March 25, 2003

LER 50-388/2000-003-00, “Inadvertent ESF Actuation Caused by Loss of RPS Power Supply,” dated August 2, 2000